

## POCONO MOUNTAINS, PA, USA

WMO: 725434

Lat: 41.139N

Lon: 75.379W

Elev: 584

StdP: 94.50

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54789

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -17.5      | -15.0 | -23.3                         | 0.5 | -15.7 | -20.9 | 0.6 | -13.6 | 12.8                  | -6.9 | 12.0 | -5.9 | 5.2                   | 300  | 0.547 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.1                   | 29.1            | 20.9 | 27.8 | 20.4 | 26.5 | 19.8 | 22.6                | 26.7 | 21.8 | 25.4 | 21.1 | 24.5 | 4.2                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.4                            | 17.2 | 24.1 | 20.9 | 16.7 | 23.6 | 20.0 | 15.8 | 22.9 | 69.4          | 26.7 | 66.5 | 25.4 | 63.9 | 24.2 | 25.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 10.8          | 9.2  | 8.2 | DB                         | -20.9 | 31.5               | 3.1 | 1.5                                                | -23.1 | 32.6       | -25.0 | 33.4       | -26.7 | 34.3       | -28.9 | 35.4 | (4) |
| (5)               |      |     | WB                         | -21.4 | 24.0               | 2.9 | 0.8                                                | -23.5 | 24.5       | -25.2 | 25.0       | -26.9 | 25.4       | -29.0 | 26.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.3         | -4.7  | -3.6  | 0.7   | 7.8   | 13.5  | 17.8  | 20.6  | 19.4  | 15.8  | 9.6   | 3.7   | -1.5  | (6)  |  |
|                                                                        | DBStd       | 10.02       | 6.30  | 5.61  | 5.80  | 5.26  | 4.60  | 3.51  | 2.72  | 2.85  | 3.66  | 4.75  | 5.05  | 5.27  | (7)  |  |
|                                                                        | HDD10.0     | 1873        | 456   | 382   | 292   | 102   | 17    | 0     | 0     | 0     | 2     | 67    | 197   | 358   | (8)  |  |
|                                                                        | HDD18.3     | 3856        | 714   | 615   | 546   | 319   | 162   | 52    | 9     | 22    | 92    | 272   | 439   | 615   | (9)  |  |
|                                                                        | CDD10.0     | 1261        | 0     | 1     | 6     | 36    | 125   | 234   | 328   | 291   | 177   | 55    | 8     | 2     | (10) |  |
|                                                                        | CDD18.3     | 201         | 0     | 0     | 0     | 2     | 12    | 36    | 79    | 54    | 17    | 2     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 1347        | 0     | 0     | 0     | 29    | 107   | 242   | 541   | 315   | 106   | 7     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 257         | 0     | 0     | 0     | 6     | 14    | 45    | 118   | 57    | 17    | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 3.8         | 4.5   | 4.7   | 4.6   | 4.3   | 3.5   | 3.2   | 2.8   | 2.6   | 2.9   | 3.7   | 4.1   | 4.5   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 1299        | 92    | 74    | 108   | 103   | 102   | 124   | 107   | 114   | 133   | 125   | 111   | 118   | (15) |  |
|                                                                        | PrecMax     | 1715        | 231   | 193   | 216   | 208   | 233   | 297   | 221   | 225   | 278   | 340   | 151   | 242   | (16) |  |
|                                                                        | PrecMin     | 995         | 46    | 28    | 29    | 54    | 23    | 55    | 42    | 34    | 43    | 31    | 54    | 51    | (17) |  |
|                                                                        | PrecStd     | 187         | 50    | 35    | 41    | 42    | 49    | 62    | 41    | 52    | 64    | 81    | 29    | 48    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 12.9  | 14.2  | 19.4  | 27.4  | 28.4  | 30.1  | 31.3  | 30.6  | 28.7  | 24.2  | 18.6  | 15.1  | (19) |  |
|                                                                        |             | MCWB        | 11.0  | 11.6  | 12.4  | 15.8  | 19.1  | 21.3  | 22.1  | 21.5  | 20.8  | 17.3  | 12.7  | 12.9  | (20) |  |
|                                                                        | 2%          | DB          | 10.0  | 10.3  | 15.3  | 22.7  | 26.3  | 27.9  | 29.3  | 28.3  | 26.5  | 21.4  | 15.9  | 11.6  | (21) |  |
|                                                                        |             | MCWB        | 8.1   | 7.8   | 10.5  | 13.3  | 18.2  | 20.4  | 21.5  | 21.0  | 20.0  | 16.2  | 12.3  | 10.0  | (22) |  |
|                                                                        | 5%          | DB          | 7.1   | 7.4   | 12.5  | 19.7  | 24.0  | 26.2  | 27.9  | 26.6  | 24.1  | 19.3  | 13.7  | 8.8   | (23) |  |
|                                                                        |             | MCWB        | 5.3   | 4.7   | 8.6   | 12.4  | 16.8  | 19.6  | 20.7  | 20.4  | 18.2  | 16.0  | 10.8  | 7.3   | (24) |  |
|                                                                        | 10%         | DB          | 4.0   | 4.2   | 9.8   | 16.7  | 21.8  | 24.2  | 26.4  | 25.0  | 22.4  | 17.3  | 11.7  | 6.3   | (25) |  |
|                                                                        |             | MCWB        | 2.3   | 1.8   | 6.3   | 10.3  | 15.5  | 18.2  | 19.9  | 19.2  | 17.7  | 14.2  | 9.2   | 4.4   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 11.9  | 12.6  | 13.6  | 17.5  | 21.0  | 22.7  | 23.8  | 23.5  | 22.5  | 19.8  | 15.4  | 13.5  | (27) |  |
|                                                                        |             | MCDB        | 12.8  | 14.3  | 17.2  | 22.7  | 26.3  | 26.9  | 28.8  | 27.7  | 26.4  | 21.5  | 17.4  | 14.5  | (28) |  |
|                                                                        | 2%          | WB          | 8.7   | 8.0   | 11.3  | 15.3  | 19.7  | 21.7  | 22.7  | 22.4  | 21.2  | 18.3  | 13.4  | 10.3  | (29) |  |
|                                                                        |             | MCDB        | 9.5   | 9.3   | 14.0  | 20.6  | 24.1  | 25.7  | 27.3  | 25.9  | 24.1  | 20.1  | 14.8  | 11.3  | (30) |  |
|                                                                        | 5%          | WB          | 5.3   | 4.7   | 9.2   | 13.5  | 18.4  | 20.7  | 21.9  | 21.6  | 20.0  | 16.6  | 11.4  | 7.6   | (31) |  |
|                                                                        |             | MCDB        | 6.6   | 6.9   | 11.8  | 17.8  | 22.2  | 24.7  | 25.6  | 24.6  | 22.5  | 18.5  | 13.0  | 8.6   | (32) |  |
|                                                                        | 10%         | WB          | 2.5   | 1.9   | 6.6   | 11.7  | 17.1  | 19.8  | 21.2  | 20.7  | 18.9  | 14.6  | 9.3   | 4.6   | (33) |  |
|                                                                        |             | MCDB        | 4.0   | 4.2   | 9.4   | 15.9  | 19.9  | 23.1  | 24.6  | 23.6  | 21.2  | 16.8  | 11.2  | 6.0   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 7.6   | 8.3   | 9.1   | 10.6  | 10.5  | 9.9   | 10.1  | 9.7   | 9.9   | 8.9   | 8.2   | 6.9   | (35) |  |
|                                                                        |             | MCDBR       | 10.3  | 11.0  | 12.4  | 14.3  | 12.8  | 11.6  | 11.5  | 10.9  | 12.2  | 11.7  | 10.7  | 9.4   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 9.4   | 9.2   | 8.4   | 7.4   | 6.3   | 5.6   | 5.1   | 5.0   | 6.3   | 6.8   | 7.7   | 8.7   | (37) |  |
|                                                                        |             | MCWBR       | 10.1  | 10.3  | 11.1  | 12.5  | 10.7  | 9.9   | 9.7   | 9.0   | 9.5   | 9.1   | 9.6   | 9.2   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.284 | 0.306 | 0.331 | 0.350 | 0.385 | 0.434 | 0.438 | 0.428 | 0.353 | 0.339 | 0.301 | 0.283 | (40) |  |
|                                                                        |             | taud        | 2.451 | 2.381 | 2.407 | 2.397 | 2.322 | 2.210 | 2.209 | 2.227 | 2.415 | 2.469 | 2.538 | 2.521 | (41) |  |
|                                                                        | Ebn at Noon |             | 879   | 906   | 920   | 921   | 891   | 845   | 836   | 834   | 883   | 857   | 855   | 855   | (42) |  |
|                                                                        |             | Edh at Noon | 77    | 96    | 105   | 114   | 126   | 142   | 140   | 134   | 103   | 87    | 71    | 67    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.70  | 2.57  | 3.52  | 4.56  | 5.15  | 5.57  | 5.77  | 5.01  | 4.14  | 2.80  | 1.92  | 1.41  | (44) |  |
|                                                                        | RadStd      |             | 0.13  | 0.30  | 0.30  | 0.48  | 0.55  | 0.51  | 0.34  | 0.32  | 0.51  | 0.33  | 0.20  | 0.16  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (58 neighbors) |  | +0.40 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +115    | +64     |

Nomenclature: See separate page